

Solving Systems Without a Graph

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. adults and children = $a = 8, c = 4$ | 7. price of each item = $b = 3, m = 4$ |
| 2. price of each item = $d = 3, h = 4$ | 8. (a) the (months, cost) pair = $x = 5, y = 90$,
— |
| 3. pens and notebooks = $n = 10, p = 20$ | 9. (a) solution, or none = $\emptyset$, — |
| 4. mass of each box size = $l = 8, s = 5$ | 10. (a) boat speed and current speed = $b = 16$,
$c = 4$, — |
| 5. (a) ticket prices = $a = 11, c = 9$, — | |
| 6. length of each piece = $x = 8, y = 18$ | |
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What is verified

10 of 14 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 5, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.C.8b – *problems 5, 7, 9*

8.EE.C.8c – *problems 1, 2, 3, 4, 6, 8, 10*

Difficulty 1–4 of 5 across 14 tagged checks.

- 1.** $a + c = 12$ and $a = c + 4$ (adults and children). Find both.

Solution: The second equation is already solved for a , so substitute it into the first: $(c + 4) + c = 12$, giving $2c = 8$ and $c = 4$. Then $a = 4 + 4 = 8$. Eight adults and four children, and $8 + 4 = 12$ checks.

$a = 8, c = 4$

- 2.** $2h + d = 11$ and $2h + 3d = 17$ (dollars). Find each price.

Solution: Both equations contain $2h$, so subtracting the first from the second removes it: $2d = 6$, so $d = 3$ dollars. Then $2h + 3 = 11$ gives $2h = 8$ and $h = 4$ dollars.

$h = 4, d = 3$

- 3.** $p + n = 30$ and $2p + 5n = 90$ (30 items, \$90 bill). Find how many of each.

Solution: Solve the first equation for p : $p = 30 - n$. Substitute into the second: $2(30 - n) + 5n = 90$, so $60 + 3n = 90$, giving $3n = 30$ and $n = 10$ notebooks. Then $p = 20$ pens. Check the bill: $2(20) + 5(10) = 90$.

$p = 20, n = 10$

- 4.** $2l + 3s = 31$ and $2l + 5s = 41$ (kilograms). Find each mass.

Solution: The $2l$ term is common, so subtract: $2s = 10$ and $s = 5$ kg. Then $2l + 15 = 31$ gives $2l = 16$ and $l = 8$ kg. $l = 8, s = 5$

5. $4a + 3c = 71$ and $2a + 3c = 49$ (dollars). (a) Both prices. (b) Why that method.

Solution (a): Subtracting the second equation from the first removes the child term: $2a = 22$, so $a = 11$ dollars. Then $2(11) + 3c = 49$ gives $3c = 27$ and $c = 9$ dollars.

$a = 11, c = 9$

Solution (b) — instructor-judged. Both equations already carry the same child term, $3c$, so one subtraction removes it and leaves a single variable. Substitution would first mean solving one equation for a variable, which here introduces thirds. Naming the matching terms as the reason elimination is quicker earns full credit.

6. $x + y = 26$ and $y = 2x + 2$ (metres). Find each piece.

Solution: The second equation is solved for y , so substitute: $x + (2x + 2) = 26$, giving $3x = 24$ and $x = 8$ metres. Then $y = 2(8) + 2 = 18$ metres, and $8 + 18 = 26$ checks.

$x = 8, y = 18$

7. $3b + 2m = 17$ and $2b + 5m = 26$ (dollars). Find each price.

Solution: Neither term matches yet, so scale first. Multiply the first equation by 5 and the second by 2:

$$15b + 10m = 85, \quad 4b + 10m = 52.$$

Subtracting removes the muffin term: $11b = 33$, so $b = 3$ dollars. Then $3(3) + 2m = 17$ gives $2m = 8$ and $m = 4$ dollars.

$b = 3, m = 4$

8. $y = 12x + 30$ and $y = 18x$ (dollars). (a) Equal-cost pair. (b) Why that method.

Solution (a): Both equations give y , so set the two expressions equal: $12x + 30 = 18x$, giving $6x = 30$ and $x = 5$ months. The cost there is $18(5) = 90$ dollars.

$(5, 90)$

Solution (b) — instructor-judged. Each equation is already solved for the same variable, so one expression can be put straight in place of the other — substitution in its quickest form, with no rearranging at all. Saying that elimination would work but needs an extra step first is also correct.

9. $2f + 3b = 46$ and $4f + 6b = 100$ (dollars per tray). (a) Solve or state none. (b) What it means.

Solution (a): Multiply the first equation by 2: $4f + 6b = 92$. The second says the same left-hand side equals 100. Since $92 \neq 100$, no pair of prices can satisfy both.

no solution

Solution (b) — instructor-judged. Doubling the first order gives exactly the second order — four trays of fruit and six of bread — so it should cost twice \$46, that is \$92. The caterer quoted \$100 for that same order, so the two quotes contradict each other and no pair of tray prices fits both. Noting that the left sides are proportional while the right sides are not is the same argument and earns full credit.

10. $b + c = 20$ and $b - c = 12$ (km per hour). (a) Both speeds. (b) Why adding works.

Solution (a): Adding the two equations cancels the current: $2b = 32$, so $b = 16$ km per hour. Then $16 + c = 20$ gives $c = 4$ km per hour. Check upstream: $16 - 4 = 12$.

$b = 16, c = 4$

Solution (b) — instructor-judged. The current appears as $+c$ in one equation and $-c$ in the other, so adding the two equations makes those terms cancel with no scaling at

all and leaves the boat's own speed in one step. Observing that subtracting instead would isolate the current's speed just as quickly is also correct.